

FOOD REMEDY API

Research: RE007

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SUMMARY

This investigation examined the current implementation of the recommendation functionality within the Food Remedy application. The application was successfully configured and tested to trace the workflow from the barcode scanning to product evaluation. The investigation section found that the personalized assessment and major recommendation section is currently implemented through For You section, while the alternative product recommendations remain partially implemented using the mock data. The report also identifies where the recommendation pipeline should integrate with the existing application and proposes how the completed recommendation system should operate.

INTRODUCTION

The aim of this research is to investigate how the recommendation functionality is currently implemented within the food remedy application and identify where it should integrate into user experience. The investigation focuses on tracing the product scanning workflow, reviewing the existing implementation, identifying incomplete or disconnected recommendation features, comparing the implementation with the previous recommendation research, and proposing how the recommendation pipeline should operate.

INITIAL INVESTIGATION

As an initial step, the application was successfully built and executed to understand the current user experience before examining recommendation

implementation. The application allowed users to create and manage dietary profiles, such as a vegan profile, vegetarian profile, add allergy which are intended to personalise the product evaluation. The main navigation consists of shopping, history, scan, settings. The testing the testing focused on the barcode scanning workflow. After scanning a supermarket product barcode, the application receives the product information and displays a product details page. Users can view nutritional information, allergen information and dietary suitability based on currently selected profile. Products can also be added directly to shopping lists, which are organised within the folders. During initial testing, no explicit recommendation section or alternative product suggestions were observed after scanning a product. Based on based on the current user interface, recommendations do not appear to be exposed to users during the product scanning process.

CURRENT PROGRESS

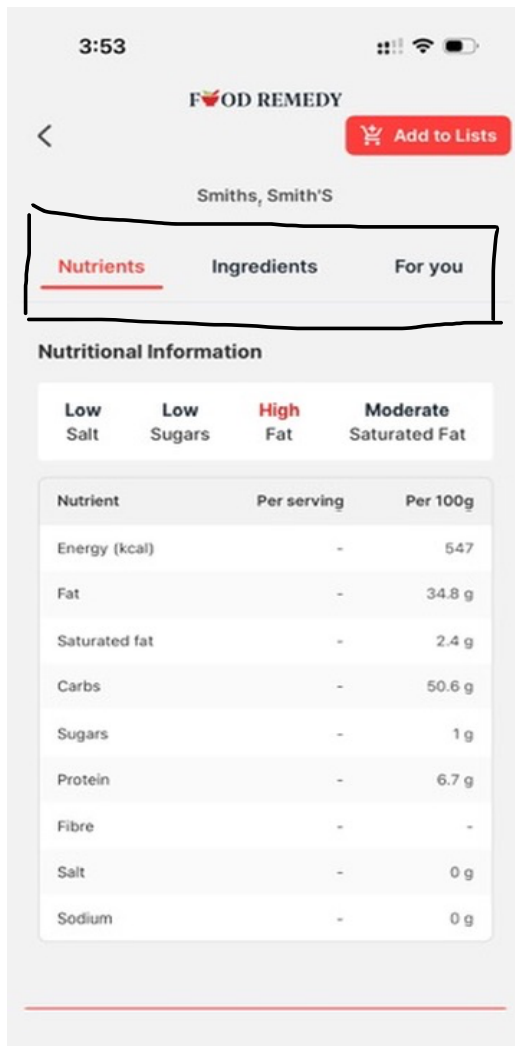
At this stage, the application has been successfully configured and executed after resolving several dependency and Expo SDK compatibility issues. Initial testing has established the end to end user workflow from product scanning to product information display. This provides the foundation required to investigate the recommendation implementation in the source code.

APPLICATION WORKFLOW:

The Food Remedy API was successfully configured and executed, allowing the recommendation workflow to be investigated. During testing, the application was explores from the product information and recommendation interfaces.

When a product barcode is scanned, the application retrieves the product information and displays several sections for the selected item. These include:

- Nutritional information (like energy, fat, carbohydrates)
- Ingredients
- For You (Personalised section)



The application also has some additional features including:

- Product scan history
- Shopping list management
- Multiple user profiles
- General settings
- Ability to add products to shopping list for future reference

EXISTING RECOMMENDATION FUNCTIONALITY:

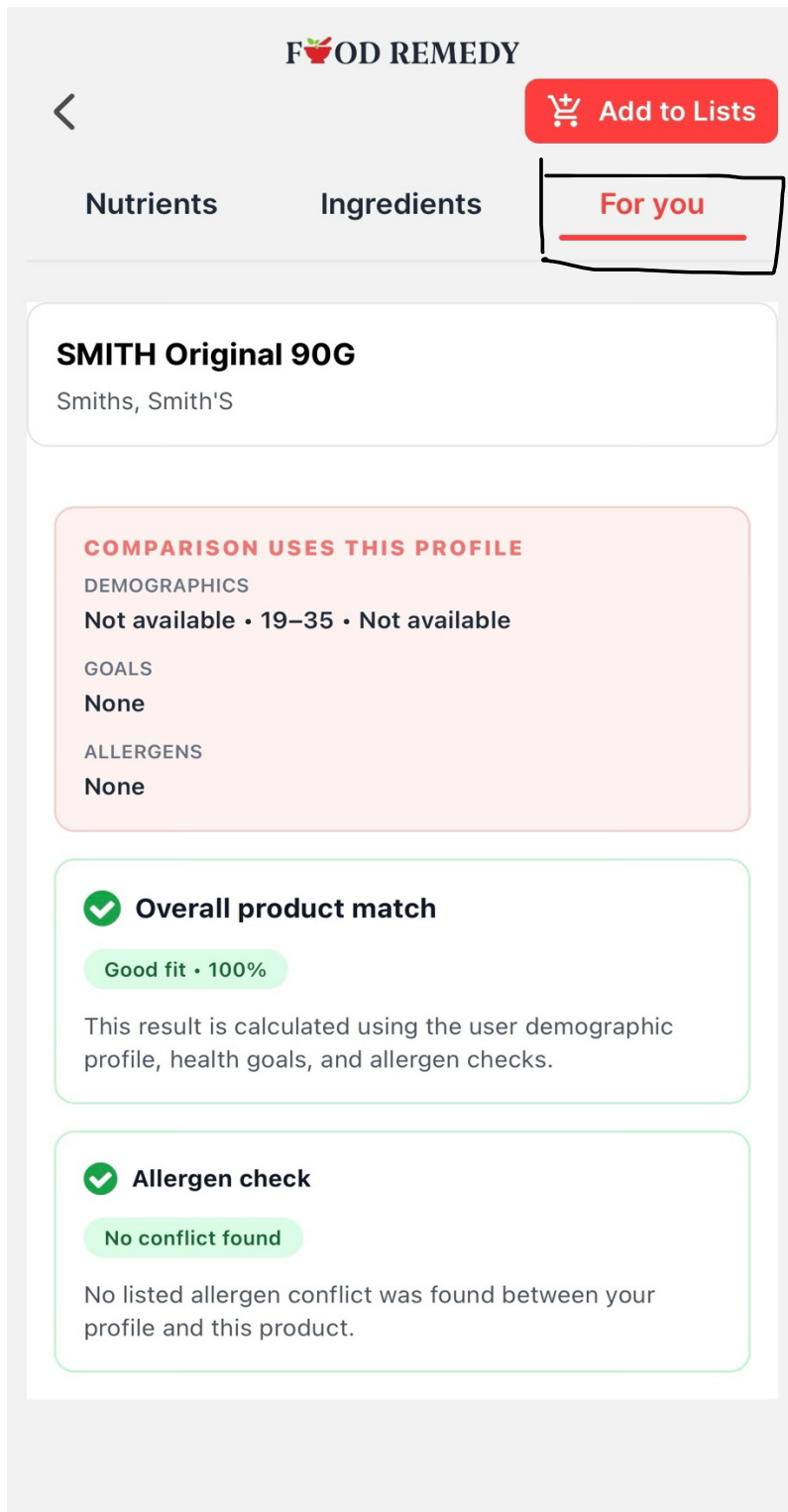
The investigation identified that the primary recommendation functionality is implemented within the FOR YOU section of the product page, Rather than simply displaying the nutritional information, this section evaluated the scanned product against current active user profile.

The personalize assessment considers information such as the user's demographic profile, dietary preferences, and allergen information. During testing, the application displayed the messages indicating that the overall product assessment was calculated using the user's profile, health goals and allergen checks. It also informed the user whether any allergen conflicts occur between the scanned product and the active profile.

RECOMMENDATION MODULE INVESTIGATION:

The recommendation module was further investigated by reviewing the implementation of the RecommendationsTab component. The codebase shows that the personalized recommendations were partially implemented. The application successfully retrieves information from the active user profile. However, inspection of the implementation of revealed that the recommended products are currently generated from a locally defined `MOCK_PRODUCTS` collection rather than a live database or recommendation service. The displayed products are therefore static demonstration data and are not dynamically selected according to the scanned product or user profile.

Even though there is a recommendation structure that includes fields for the user goals and allergens, these values are currently returned as empty arrays, indicating that this functionality has not yet been fully integrated into the recommendation engine completely.



RECOMMENDATION SERVICES AND CANDIDATE RETRIEVAL:

The recommendation implementation was further investigated by reviewing the RecommendationsTab component. The source code shows that a recommendation framework has already been developed. It retrieves information from the active user profile, including demographic attributes such

as gender, age group and activity level, which are intended to support personalized recommendations. However, the investigation found that recommendation candidates are currently retrieved from a locally defined Mock products collection rather than a backend recommendation service or a live product database. Furthermore, the recommendation interface is not currently connected to the application's visible user workflow. During testing, scanning a product only displayed the nutritional information, ingredients and For You sections, while no alternative product recommendations were presented to the user. This suggests that the recommendation feature has not yet been integrated into the production interface.

RECOMMENDATION FEATURE STATUS:

Based on the investigation, the recommendation functionality can be divided into implemented and incomplete components.

The following features are currently operational:

- barcode scanning and product retrieval
- nutritional information display
- ingredients display
- personalized assessment using active user profile in the for you section
- shopping list management
- product history scan
- multiple user profiles

The following features are still incomplete or not integrated:

- Display of alternative recommended products
- dynamic recommendation generation
- backend recommendation service
- live candidate retrieval based on scanned products
- integration of recommendation interface into the application workflow

COMPARISON WITH EXISTING RECOMMENDATION RESEARCH:

The current implementation was compared with the existing recommendation and alternative mapping research to determine how close the application aligns with the proposed recommendation pipeline. The previous research suggested that the recommendation system should analyse the scanned product together

with the users dietary profile health calls and allergen information to provide suitable alternative products. The current implementation only partially reflects this research. The application successfully performs personalized product assessment through the for you section by comparing the scanned product against the active user profile. However, the investigation found that the alternative recommendation functionality has not been fully integrated into the application. Although the source code contains a recommendation module with the mock products, these recommendations are not currently retrieved from a backend service or displayed within the application interface. Therefore, the existing implementation provides the foundation for the recommendation system but does not yet achieve the complete functionality proposed in a previous research.

IMPLEMENTATION PROPOSAL:

Based on the investigation, the recommendation pipeline should be integrated immediately after a product barcode has been scanned and the product information has been retrieved successfully. The scanned product details together with the active user profile should be passed to a recommendation service capable of analyzing the nutritional values, dietary preferences, allergens and health goals. The recommendation service should also be able to get suitable alternative products from the product database and rank them according to their compatibility with the users profile and return the highest matching products to the application interface. These recommendations should be then displayed alongside the existing personalized section on the products details page, allowing users to compare healthier or more suitable alternatives without interrupting the existing scanning workflow. The recommended products should also support the existing shopping list functionality so users can directly add recommended items to the shopping list.